

Design a Unique ID Generator in Distributed Systems

1. Problem Overview

In this problem, we need to design a **unique ID generator** for distributed systems.

2. Initial Thought (Naive Approach)

- Use **primary key with auto_increment** in a traditional database

Limitation

- Does **not work in distributed systems** because:
 - A single database server is not sufficient for scale
 - Multiple databases cannot easily coordinate to generate unique IDs
 - Ensuring uniqueness with **low latency** is challenging

3. Core Challenge

Generate **unique IDs across multiple machines** with:

- High scalability
- Low latency
- No collisions

4. Examples of Unique IDs

user_id
1227238262110117894
1241107244890099715
1243643959492173824
1247686501489692673
1567981766075453440